THE INVESTMENT BANK THAT DISCOVERS HIDDEN OPPORTUNITIES

Corporate Knowledge Graphs, Strategic Traversal and Adaptive Agentic M&A Discovery

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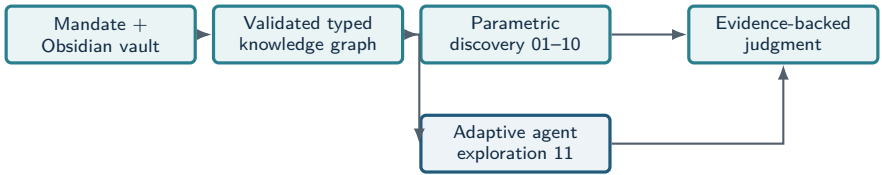
The core proposition

Acquisition value is relational—not merely classificatory.

The Second Brain links capabilities, ownership, licenses, markets and counterparties to explain hidden acquisition opportunities.

Research question Can typed corporate knowledge and constrained graph traversal uncover viable strategic acquisition hypotheses beyond conventional financial screening?

The architecture: two journeys through the same vault



Comparison The same knowledge base supports fixed search rules or LLM-directed research under deterministic governance.

Cell 01 | Runtime, configuration and reproducibility

ESTABLISH THE GOVERNED COLAB ENVIRONMENT

- Mount Google Drive and establish the exact Topic 793 working root.
- Create persistent paths for data, vault, graphs, evaluation and reports.
- Seed the synthetic experiment and configure Claude Haiku 5.5 through Colab Secrets.
- Distinguish deterministic execution from actual model-generated judgment.

Key lesson Reliable discovery begins with reproducible computation and explicit API status.

Cell 02 | The acquisition mandate and synthetic corporate universe

DEFINE THE INVESTMENT BANKING RESEARCH PROBLEM

- Aurora Financial Group seeks cross-border payments, real-time settlement and digital identity.
- Create 2,000 synthetic firms across 25 jurisdictions and eight sectors.
- Assign financial profiles, primary and secondary capabilities, licensing and control indicators.
- Plant five strategic benchmark opportunities and three deceptive ineligible candidates.

Key lesson The synthetic universe includes hidden opportunities and deliberately attractive traps.

Cell 03 | Obsidian vault and typed corporate relationships

REPRESENT CORPORATE KNOWLEDGE AS NAVIGABLE EVIDENCE

- Write Markdown notes for companies, owners, capabilities, licenses, markets and sources.
- Build explicit relations: controls, possesses capability, holds license, serves market, distributes for and depends on.
- Preserve relation sources, dates, confidence and readable backlinks.
- Export node and edge ledgers alongside the human-readable Obsidian vault.

Key lesson The vault preserves knowledge; typed edges make its relationships computationally actionable.

Cell 04 | Validated master graph and five derived views

TURN SOURCE RECORDS INTO CONTROLLED GRAPH PROJECTIONS

- Validate company IDs, relationship endpoints, source IDs, confidence and ownership percentages.
- Create a directed NetworkX multigraph preserving typed edges and metadata.
- Project ownership, capability, regulatory, commercial and diligence relationship views.
- Export graph summaries and dedicated edge ledgers for downstream analysis.

Key lesson Graph views are decision-specific projections of one underlying evidence system.

Cell 05 | Question-driven multi-hop traversal

CONNECT AURORA STRATEGIC GAPS TO CANDIDATES

- Start from Aurora strategic gaps and match them against available corporate capabilities.
- Trace candidate licenses, markets, distribution relationships and dependencies.
- Explicitly label GAP-to-CAP alignment as a derived match, not a sourced vault edge.
- Preserve illustrative paths, eligibility indicators and outgoing relation context.

Key lesson A traversal path is an inspectable acquisition hypothesis—not a proof of feasibility.

Cell 06 | Four discovery engines on one universe

BENCHMARK RETRIEVAL MECHANISMS UNDER IDENTICAL FACTS

- Tabular method ranks growth, EBITDA margin and valuation multiple.
- Semantic method uses TF–IDF and cosine similarity of company descriptions.
- Graph method rewards strategic gap matches and selected network signals.
- Hybrid combines 65 percent graph with 35 percent semantic signal, then applies hard eligibility gates.
- Compare planted-target precision and recall among the top twenty candidates.

Key lesson Graph and hybrid improve planted-opportunity recall in this synthetic design; not validated market efficacy.

Cell 07 | Investment committee evidence packages

MOVE FROM RANKED TARGETS TO EVIDENCE-BACKED CONTEXT

- Select the top three eligible candidates from the hybrid screening results.
- For each target assemble financial profile, strategic gaps, controls, license and review indicators.
- Collect relevant relationship edges, paths and admissible source identifiers.
- Generate a structured JSON dossier with explicit due-diligence questions.

Key lesson The LLM receives an evidence package, not an unsupported company recommendation.

Cell 08 | Bounded Claude investment judgment

INTERPRET EACH CANDIDATE SEPARATELY

- Send one evidence package at a time to Claude Haiku 5.5.
- Request compact JSON containing thesis, transaction structure, synergies, risks and diligence.
- Capture token usage, provider stop reason, raw text and parse status.
- Never treat a truncated or unparseable response as an investment judgment.

Key lesson The model interprets evidence; its narrative does not establish regulatory approval.

Cell 09 | Validation and adverse-fact counterfactual

GOVERN CLAIMS AND TEST RECOMMENDATION SENSITIVITY

- Check parsed assessments, candidate identities and admissible citation IDs.
- Confirm shortlisted companies satisfy licensing and control availability filters.
- Introduce a synthetic new fact: the initial leading target has no available control rights.
- Recompute the shortlist under the unchanged hybrid score and revised eligibility.

Key lesson An adverse fact must override a favorable strategic score.

Cell 10 | Final reproducible laboratory report

CONSOLIDATE THE OUTPUT WITHOUT OVERSTATING RESULTS

- Export strategic mandate, graph scale and the four-method synthetic benchmark.
- Describe shortlisted acquisition hypotheses and document the live Claude status truthfully.
- Include counterfactual changes, validation findings and an explicit governance conclusion.
- Persist reports, evaluation tables, source graph ledgers and portable vault notes.

Key lesson The output is a research memorandum with evidence and limits—not transaction approval.

Cell 11 | Adaptive agentic discovery: a second journey

TRANSFER TRAVERSAL PLANNING FROM PYTHON TO A BOUNDED LLM AGENT

- Parametric path: fixed traversal and ranking, followed by downstream model judgment.
- Agentic path: Claude formulates hypotheses, chooses graph tools and revises the search.
- Allowed tools include company search, node expansion, path finding and eligibility evaluation.
- Python enforces schema, provenance, tool budget and hard constraints independently.
- Evaluate discovery gain, false positives, cost and reproducibility in controlled repeated runs.

Key lesson The LLM proposes where to look next; the graph determines which facts actually exist.

Why graphs rather than only relational tables?

MODELING CHOICE, NOT A CLAIM OF IMPOSSIBILITY

- Normalized SQL tables can represent the same entities and relationships, including recursive joins.
- Typed graph paths simplify multi-hop exploration across heterogeneous relationship classes.
- Graph navigation exposes intermediate entities and supports provenance-oriented explanations.
- Neither graphs nor LLMs guarantee faster retrieval, better facts or economically viable deals.

Key lesson The benefit is explainable relationship-centered exploration, not a magical replacement for SQL.

The evaluation question for Cell 11

WHAT WOULD CONSTITUTE MEASURED AGENTIC VALUE?

- Run parametric and agentic algorithms against the identical vault, truth labels and eligibility constraints.
- Measure planted-opportunity recall, novel valid discoveries, false positives and evidence coverage.
- Record model calls, tokens, latency, tool attempts, repeats and path provenance.
- Treat illustrative iteration examples as hypotheses, not observed performance.

Key lesson The next contribution is a controlled comparison of discovery quality per unit of research cost.

Institutional lessons and limitations

RESPONSIBLE DEPLOYMENT OF THE INVESTMENT BANKING SECOND BRAIN

- Structured knowledge enables auditable searches across complex financial institutions.
- Hard constraints and primary diligence remain separate from strategic screening scores.
- Agentic autonomy belongs in hypothesis and query planning—not in fact fabrication.
- Human bankers remain responsible for valuation, execution, licensing and committee decisions.

Key lesson Represent knowledge, navigate evidence, exercise judgment, validate conclusions.

SECOND BRAIN | LABORATORY 03

FROM CORPORATE INFORMATION TO STRATEGIC DISCOVERY

Knowledge vaults + typed graphs + parametric traversal
Adaptive agentic exploration + governed investment judgment

The Second Brain Project | Investment Banking Laboratory